

Refugee Inflow and Native Parity Progression

Evidence from Turkey

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1. Turkey hosts the largest refugee population in the world.

- About 3.6 million Syrians by 2018, roughly 4.4 percent of the population.

2. Refugees have much higher fertility than natives.

- Syrian refugee's total fertility rate (TFR) is about 5.3 much higher than Turkey's 1.51.

⇒ **An ideal setting** to examine the interaction of refugee inflows and native fertility decisions.

Research question. *How does the influx of Syrian refugees affect the fertility decisions of native Turkish women?*

I study annual birth transitions (does the n th child arrive this year?) and cumulative parity by age, for parities 1 to 5.

Turkish vs. Syrian TFR

Descriptives

- Using the 2013 and 2018 rounds of the Turkish Demographic and Health Survey (TDHS), I reconstruct a woman-year pseudo-panel from age 12 to the survey date.
- I merge province-year Syrian refugee-to-native ratios.
 - Each woman's exposure is assigned at her province of residence in 2011, the last pre-inflow year, reconstructed from her migration history.
- I estimate both discrete-time hazard and cumulative parti^y attainment models of marriage and birth transitions.
- To address endogenous refugee settlement, I instrument exposure using a distance-based shift-share IV.
- Because I test many related outcomes, I report false-discovery-rate q -values.

Why I Expect an Effect

1. Labor markets and income

- Lower female employment with stable male employment lowers the opportunity cost of children without an income loss (Aksu, Erzan, and Kırdar 2022; Demirci 2023).

2. Childcare and household services

- In developed hosts, immigration lowers care prices and raises fertility of educated women (Furtado and Hock 2010; Furtado 2016; Romiti 2018).

3. Housing costs and crowding

- Higher rents and living costs can delay marriage and childbearing (Balkan and Tumen 2016; Akgündüz, van den Berg, and Hassink 2018).

4. Cost of child rearing: schooling and health

- School congestion and switching to private schools raise the cost per child (Tumen 2019; Kırdar, Dayıoğlu-Tayfur, and Koç 2022); congested maternal and child health services change the perceived cost or risk of childbearing (Aygün et al. 2021; Erten and Keskin 2022).

5. Social norms and cultural proximity

- Most refugees are Arabs and Kurds from northern Syria, sharing language and family norms with the native minority in the most exposed provinces.

⇒ Different channels working in different directions.

- **Exposure accelerates the transition to a third birth.** A 10 percentage point increase in the refugee-to-native ratio raises the annual probability of a third birth by 3.1 percentage points
 - This is a 38 percent increase from a baseline of 8.2 percent.
- **No other birth transition responds.**
- **Tempo not level effects.** Cumulative parity by fixed ages shows no significant effect: exposure moved the timing of third births; higher completed fertility is not yet visible at the observed ages.
- **The response is present where the refugees actually settled:** precisely estimated within the top exposure tercile and quintile.
- **Suggestive cultural proximity gradient:** Kurdish and Arabic speaking women show larger responses across designs and in stated ideal family size.
- **Mechanism:** consistent with gender-asymmetric labor market adjustment; female employment falls where fertility rises, husbands' employment is stable.

Main source: 2013 and 2018 rounds of the TDHS, nationally representative microdata on women with complete birth histories and migration histories.

- **Pseudo-panel:** one observation per woman per year from age 12 to the survey date; these are the only rounds with birth and migration histories for all women, ever and never married.
- **Panel window:** 1987 to 2018, ages 12 to 36. In 1987 the oldest observed cohort (age 36 in 2011) was age 12, so the panel holds one complete pre-inflow risk horizon.
- **Outcome:** $h_{it}^p = \mathbf{1}\{p\text{-th birth in year } t\}$, on the risk set of women with exactly $p - 1$ children.
- **Exposure:** the refugee-to-native ratio R_{jt} at the woman's 2011 province.

Why only 2013 and 2018?

Why 1987: start-year sensitivity

Turkish vs. Syrian descriptives

By age group

Data: Reconstructing the Pseudo-Panel

Cross-sectional record (TDHS interview):

ID	Birth Yr	Childhood Prov	Move Yr	Dest. Prov	Birth 1	Birth 2
10001	1990	Adana	2009	Mardin	2010	2013

⇓ expanded into a woman-by-year panel ⇓

Pseudo-panel:

ID	Age	Year	Province	Gave Birth 1	Gave Birth 2
10001	12	2002	Adana	0	0
⋮	⋮	⋮	⋮	⋮	⋮
10001	20	2010	Mardin	1	0
10001	23	2013	Istanbul	1	1

Each birth event is placed in its recorded calendar year. The hazard design uses the woman-year rows directly; the cumulative design collapses to one row per woman per age cutoff.

Challenge: refugee location choice is endogenous. Refugees moved toward provinces with better labor markets, so OLS confounds exposure with local conditions that also affect fertility.

Quasi-exogenous variation from geography. Early inflows were shaped by border entry and camps near Syria, creating a persistent **distance gradient** that still predicts exposure as refugees dispersed west.

Following Aksu, Erzan, and Kırdar (2022), I instrument R_{jt} with

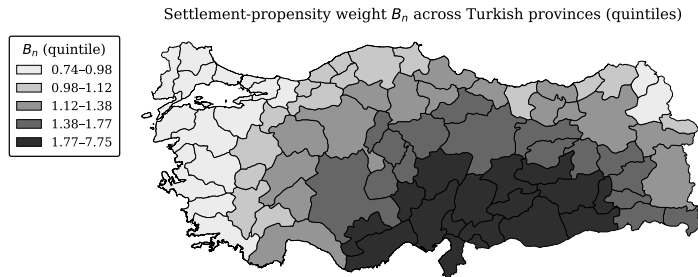
$$Z_{n,t} = \sum_{s=1}^{13} \left[\frac{\left(\frac{1}{d_{s,T}} \right) \pi_s}{\sum_{X \in \{T,L,J,I\}} \left(\frac{1}{d_{s,X}} \right)} \right] \frac{T_t}{d_{n,s}} = B_n \times T_t,$$

where π_s is the pre-war population share of Syrian province s , d are distances to Turkish provinces and to the alternative hosts (Lebanon, Jordan, Iraq), and T_t is the total Syrian refugee stock across the four host countries.

Notation in full

Dispersion maps

The Instrument: a Distance-Based Settlement Propensity



The time-invariant geographic component B_n , activated over time by the aggregate stock:
 $Z_{n,t} = B_n \times T_t$. First-stage F : 53 to 172.

Full formula

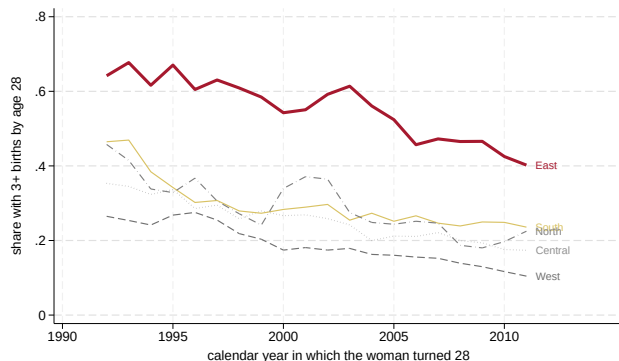
First-stage estimates

$$h_{ijt}^p = \alpha + \beta_p R_{jt} + \mathbf{X}_i' \Gamma + \delta_j + \mu_t + \alpha_a + \theta_{n(j)} t + \lambda_{m(i)} t + \varepsilon_{ijt}$$

- Separate 2SLS regressions for each birth transition $p \in \{1, \dots, 5\}$, ages 12 to 36.
- h_{ijt}^p : woman i in province j has her p -th birth in year t ; the risk set is women with exactly $p - 1$ children.
- R_{jt} : refugee-to-native ratio in province j in year t , instrumented by $Z_{jt} = B_j \times T_t$.
 - β_p is the unit effect of moving R_{jt} from 0 to 1: a 0.10 increase in R_{jt} implies a $0.10\beta_p$ percentage-point change in the annual probability of the p -th birth.
- $\mathbf{X}_i'$: survey year, education, childhood province, mother tongue, childhood residence type.
- $\delta_j, \mu_t, \alpha_a$: province, year, and age fixed effects.
- $\theta_{n(j)} t, \lambda_{m(i)} t$: NUTS-2 and mother-tongue specific linear trends.
- DHS sampling weights; standard errors clustered by province.

Transition probabilities by age

Fertility Was Trending Differently Across Regions



Refugees settled where fertility was highest and trending differently. This motivates the NUTS-2 and mother-tongue specific trends; results are not sensitive to this choice.

Trend differences across NUTS-2 regions

Plausibility of the Identification Assumption

Before 2012 exposure is essentially zero, so the settlement weight should predict nothing in the pre-period. On **pre-2012 person-years** I estimate

$$h_{ijt}^p = \gamma_p (B_j \times t) + \mathbf{X}_i' \Gamma + \delta_j + \mu_t + \alpha_a + \theta_{n(j)} t + \lambda_{m(i)} t + \varepsilon_{ijt},$$

where $\gamma_p = 0$ under the identifying assumption.

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
$B_j \times \text{trend}$	0.00029 (0.00022)	-0.00019 (0.00033)	-0.00047 (0.00035)	0.00119 (0.00081)	0.00110 (0.00079)
Unadjusted p -value	0.207	0.570	0.185	0.147	0.170
BH q -value	[0.259]	[0.570]	[0.259]	[0.259]	[0.259]
Person-years	139,819	42,159	40,882	19,752	8,652

No transition rejects, unadjusted or corrected; the cumulative design's 26-cell placebo grid is equally clean.

What is the BH correction?

Cumulative placebo

Placebos under other trends

Main Result: the Third Birth Responds

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
Refugee-to-native ratio R_{jt}	-0.070 (0.076)	0.235 (0.147)	0.313*** (0.106)	0.066 (0.099)	-0.048 (0.096)
BH q -value	[0.596]	[0.274]	[0.015]	[0.616]	[0.616]
Baseline transition probability	0.062	0.160	0.082	0.081	0.097
First-stage F	53.5	98.1	118.4	172.1	64.6
Person-years	163,980	51,359	51,677	25,161	10,806

Only the third birth survives the correction: **0.313 percentage points per percentage point of exposure** on an 8.2 percent baseline, so a 10 point exposure difference raises the annual third-birth probability by about 38 percent. Cells show 2SLS coefficients, clustered SEs, and BH q -values within the five-transition family.

[Result graphically](#)[Why not marriage?](#)[Age-range trends](#)[Lagged exposure](#)[Other trend specifications](#)[OLS and first stage](#)[Start year](#)[Residence definition](#)[Ever-married samples](#)

Is the Hazard Enough?

- The hazard identifies the **flow**: third births arrive sooner where exposure is higher.
- It cannot distinguish **tempo** (the same third child, earlier) from **quantum** (more third children ever).
- To separate them, examine the **stock**: one observation per woman per age cutoff,

$$Y_{ija}^{(p)} = \alpha + \beta^{(p,a)} \bar{R}_{ja} + \mathbf{X}_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{n(j)} t(a, i) + \lambda_{m(i)} t(a, i) + \varepsilon_{ija}$$

- **Outcome**: $Y_{ija}^{(p)} = \mathbf{1}\{\text{parity} \geq p \text{ by age } a\}$, for $a \in \{16, 20, 24, 28, 32, 36\}$; 26 estimable parity-age cells. The outcome is **absorbing**: unlike the hazard, where a woman exits the risk set at the event, here Y switches from 0 to 1 at the p -th birth and stays 1 at every later age.
- **Exposure**: the average over the woman's risk window, $\bar{R}_{ja} = \frac{1}{a-11} \sum_{s=12}^a R_{j,t(s)}$, instrumented by the identically averaged shift-share $\bar{Z}_{ja} = B_j \bar{T}_a$.
- Controls, fixed effects, and trends as in the hazard design; $\mu_{t(a,i)}$ is now a fixed effect for the calendar year in which woman i reaches age a .

Graphical intuition

The Stock: Cumulative Parity by Age

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	-0.11 (0.18)	-0.26 (0.77)	-1.12 (2.51)	0.02 (1.44)	-2.17 (1.82)	-0.73 (1.58)
Parity ≥ 2	0.02 (0.04)	-0.10 (0.46)	-0.81 (1.01)	-1.06 (2.56)	-0.48 (3.48)	-2.47 (1.58)
Parity ≥ 3	0.00 (0.01)	0.30 (0.21)	-0.76 (0.76)	0.74 (1.76)	1.41 (2.81)	1.71 (1.96)
Parity ≥ 4	–	–	0.24 (0.41)	1.63** (0.83)	0.78 (2.43)	3.36 (2.64)
Parity ≥ 5	–	–	0.23 (0.23)	1.30 (0.87)	1.34 (1.19)	3.52 (3.19)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

One cell in 26 is significant unadjusted and none survives the correction; the matching placebo grid is clean.

The stock has not moved: what the hazard picks up is **tempo**. Quantum is not yet visible at the observed ages, and cannot yet be, because the exposed cohorts have not finished childbearing.

OLS counterpart

Other trend specifications

Their placebos

Heterogeneity: Estimates Within Subsamples

Subgroup	First birth	Second birth	Third birth	Fourth birth	Fifth birth
Turkish mother tongue	-0.120* (0.068)	0.199 (0.226)	0.220* (0.121)	0.094 (0.107)	0.025 (0.153)
Kurdish/Arabic mother tongue	-0.105 (0.114)	0.512* (0.296)	0.504** (0.256)	0.129 (0.153)	0.007 (0.230)
Top exposure tercile	-0.153 (0.094)	0.007 (0.140)	0.396*** (0.110)	0.146 (0.136)	-0.117 (0.123)
Top exposure quintile	-0.161 (0.105)	0.017 (0.170)	0.392*** (0.140)	0.135 (0.162)	-0.250** (0.124)

Separate hazard regressions within each subsample. Bold: survives the BH correction within that subsample's five transitions ($q < 0.05$). The **tercile** and **quintile** subsamples pass their own pre-period tests, so the response is present and precise where the refugees actually settled; the Kurdish/Arabic subsample does not, which is taken up on the next slide.

All subgroups

Subsample pre-period tests

Interaction tests

OLS and other trends

Most refugees are Arabs and Kurds from adjacent northern Syria, closest in language and family norms to this native minority.

- Their third-birth hazard estimate is more than twice the Turkish-speaking estimate: **0.504 versus 0.220**.
- Their cumulative parity 3 by age 28 cell survives its own family's correction ($q = 0.014$).
- Their stated ideal family size shifts most at the four and five child thresholds.

I present this as **suggestive**: the subsample shows differential pre-trends, and no between-group difference survives correction in pooled interaction tests.

K/A cumulative grid

Interaction tests

Few-cluster inference

Mechanism: Gender-Asymmetric Labor Market Adjustment

Cross-sectional 2SLS: does the woman work, and did her husband work last week, on average exposure $\bar{R}_j$ at her province, instrumented by the averaged shift-share.

	All women	Kurdish/ Arabic	Lower education	Border region
<i>Panel A: Woman currently working</i>				
Average refugee-to-native ratio $\bar{R}$	-2.651** (1.029)	-3.944** (1.674)	-4.790*** (1.268)	-2.199* (1.223)
Observations	4,126	936	3,049	1,096
<i>Panel B: Husband worked in the last seven days</i>				
Average refugee-to-native ratio $\bar{R}$	2.013* (1.041)	-0.611 (1.482)	2.768** (1.250)	-0.814 (0.549)
Observations	12,760	2,869	10,076	3,659

Women's employment falls where fertility rises; husbands' employment is stable: the opportunity cost of a mother's time fell while household income held up. This is the pattern the opportunity-cost channel predicts, though these regressions alone cannot rule out other channels.

Specification details

OLS and other trends

Mechanism: Stated Ideal Family Size

If the shadow price of children fell, stated preferences should move with behavior:

Subgroup	Primary	Threshold decomposition			
	Ideal n	Ideal ≥ 2	Ideal ≥ 3	Ideal ≥ 4	Ideal ≥ 5
Full sample (pooled)	1.197** (0.541)	0.046 (0.061)	0.174 (0.254)	0.514* (0.270)	0.256** (0.109)
Lower education	1.533*** (0.549)	0.001 (0.088)	0.203 (0.247)	0.676*** (0.247)	0.351*** (0.101)
Arabic/Kurdish mother tongue	3.034** (1.253)	0.129 (0.158)	0.512 (0.417)	0.911*** (0.350)	0.475*** (0.184)
High-exposure border region	1.566** (0.734)	0.069 (0.117)	0.212 (0.237)	0.727** (0.336)	0.368** (0.143)

- A 10 point increase in exposure raises stated ideal family size by about 0.12 children, loading at the **four and five child thresholds**.
- Strongest for lower-education and Kurdish/Arabic women, the same groups as the fertility estimates. Ever-married sample of all four rounds.

⇒ **How this fits the story**: a cheaper marginal child shows up first in what women say they want and in *when* the next child arrives, and only later in completed families — the sequence behind tempo now, quantum not yet.

OLS and other trends

AAO (CESifo WP 11683) study the same question on the **NSDVW 2008 and 2014** ever-married samples with a distance IV: exposure raises the probability of a past-year birth (0.318**), concentrated among existing mothers.

I replicated the paper in full. Its sample inherits the Ertan–Keskin cleaning pipeline: about 1,900 women are dropped non-randomly (birth-date missingness, an asymmetric 2008 schooling recode) — the dropped women are older, less educated, more Kurdish, more rural, with about 1.3 more children.

Outcome	Paper IV	Faithful replication	+ Sample fixes	+ Region×year trends
Gave birth in past year	0.318**	0.333	0.192	-0.256
Birth and not first child	0.305***	0.362**	0.262*	-0.044
Number of children	1.870**	2.466*	2.225*	3.046***
Currently pregnant	0.248**	0.414**	0.378*	0.366
Pregnant and ≥ 1 child	0.317**	0.329	0.349**	0.384**

Restoring the improperly dropped women strips the headline of significance; adding region-by-year controls — the analogue of my trend battery — flips its sign. The completed-fertility and existing-mother margins survive.

[Audit details](#)

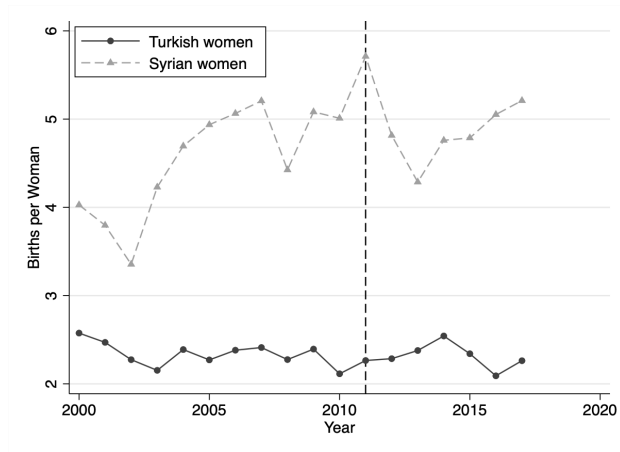
- **Independent check on my data.** Running AAO's exact design on TDHS 2008 and 2013: the past-year-birth headline is 0.181 and insignificant (-0.073 with region-by-year controls); their number-of-children result survives (3.51^{**}).
- **The surviving evidence is the existing-mother margin** — more births among women who already have children — which is qualitatively consistent with my third-birth transition result.
- **Their fragile headline is what my design guards against.** The AAO estimate has no pre-inflow placebos, no differential regional trends, and no multiple-testing correction; exactly those three checks are what their result fails and mine survives.
- **Different margins, no contradiction.** Their pooled annual-birth indicator cannot separate parities or tempo from quantum; my design locates the response at the third birth and identifies it as timing.

- **One precise result:** refugee exposure accelerated the transition to the third birth, 0.313 percentage points per point of exposure ($q = 0.015$), robust to every trend specification, start year, residence definition, and to lagging exposure to the conception year.
- **One disciplined null:** no cumulative parity cell survives correction despite clean placebos, so the evidence is about the timing of births, not yet completed family size.
- **Where and for whom:** precise within the most exposed provinces; suggestive cultural proximity gradient for Kurdish and Arabic speaking women.
- **Why:** the pattern fits a lower opportunity cost of childbearing from gender-asymmetric labor market adjustment, supported by employment and stated-preference evidence.
- **Next:** a future TDHS round will settle whether faster progression raises completed fertility or compresses birth spacing.

Thank you.

Appendix

Appendix: Total Fertility Rates, Turkish and Syrian Women



TDHS birth histories; the Syrian series uses the 2018 round. The dashed line marks 2011.

[◀ Back](#)

Appendix: Fertility Outcomes, Turkish vs. Syrian Populations

Variable	Turkish			Syrians		
	Mean	SD	Obs	Mean	SD	Obs
Age at 1st Marriage	20.54	4.23	12,320	18.52	4.09	1,849
Age at 1st Birth	22.06	4.39	11,354	19.98	3.99	1,684
Age at 2nd Birth	24.88	4.64	9,068	21.80	3.92	1,393
Age at 3rd Birth	27.11	4.83	4,938	24.05	3.88	1,050
Age at 4th Birth	28.07	4.76	2,359	26.39	4.09	750
Age at 5th Birth	29.09	4.58	1,211	28.48	4.17	506

[◀ Back](#)

Appendix: Fertility and Marriage by Age Group

	Number of Children		Portion Married	Obs
Age Group	Mean	Median		
Turkish Women				
12-16	0.001	0	0.010	967
17-21	0.175	0	0.214	2,441
22-26	0.904	1	0.631	2,375
27-30	1.671	2	0.878	1,974
31-35	2.306	2	0.940	2,681
36-40	2.827	2	0.960	2,521
41-45	2.952	3	0.964	2,157
46-49	3.100	3	0.963	1,594
Total	1.825	2	0.737	16,710
Syrian Women				
12-16	0.109	0	0.212	156
17-21	0.927	1	0.754	480
22-26	1.913	2	0.901	426
27-30	2.941	3	0.928	320
31-35	3.735	4	0.927	317
36-40	4.888	5	0.926	242
41-45	5.418	5	0.935	184
46-49	6.011	6	0.912	91
Total	2.766	2	0.834	2,216

$$Z_{n,t} = \sum_{s=1}^{13} \left[\frac{\left(\frac{1}{d_{s,T}} \right) \pi_s}{\sum_{X \in \{T,L,J,I\}} \left(\frac{1}{d_{s,X}} \right)} \right] \frac{T_t}{d_{n,s}}$$

- $s \in \{1, \dots, 13\}$: Syrian origin provinces; π_s : pre-war population share of province s .
- $X \in \{T, L, J, I\}$: host countries Turkey, Lebanon, Jordan, Iraq; $d_{s,X}$: minimum distance from s to host X .
- T_t : total Syrian refugee stock across the four hosts in year t , exogenous to Turkish conditions.
- $d_{n,s}$: distance from Turkish province n to Syrian province s .
- The bracket is the geographic probability that a refugee from s chooses Turkey; distance then distributes the predicted inflow across Turkish provinces: $Z_{n,t} = B_n \times T_t$.

◀ Back

Appendix: Refugee Dispersion Over Time

2013

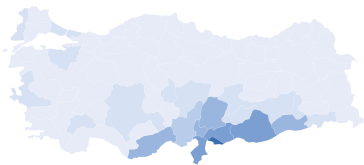
Percentage of Syrian Refugees to Natives (2013)



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2015

Percentage of Syrian Refugees to Natives (2015)



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2018

Percentage of Syrian Refugees to Natives (2018)



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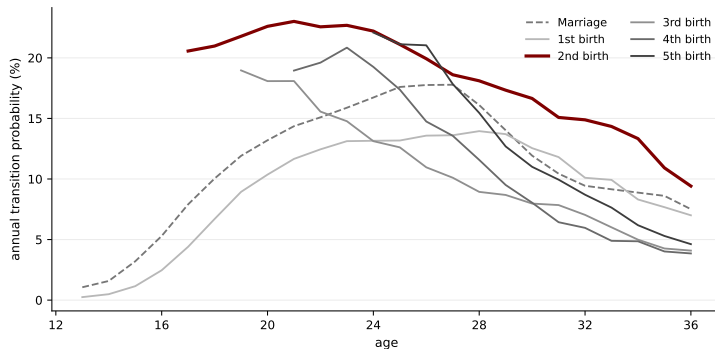
Registered Syrians over native population by province.

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
$Z_{jt} (\times 10^{-5})$	2.579*** (0.353)	2.313*** (0.234)	2.296*** (0.211)	2.463*** (0.188)	2.802*** (0.349)
First-stage F	53.5	98.1	118.4	172.1	64.6
Person-years	163,980	51,359	51,677	25,161	10,806

First stage of the hazard design: R_{jt} on $Z_{jt} = B_j \times T_t$, estimated on each transition's risk set with the full set of controls, fixed effects, and trends. The instrument is strong on every risk set.

[◀ Back](#)

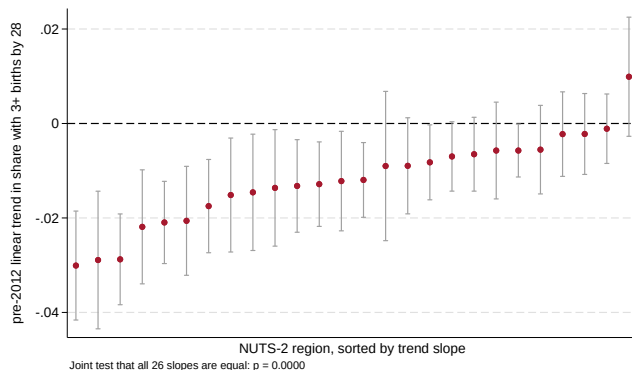
Appendix: Annual Transition Probabilities by Age



Weighted share of at-risk women making each transition at each age, TDHS 2013 and 2018 person-years.

[◀ Back](#)

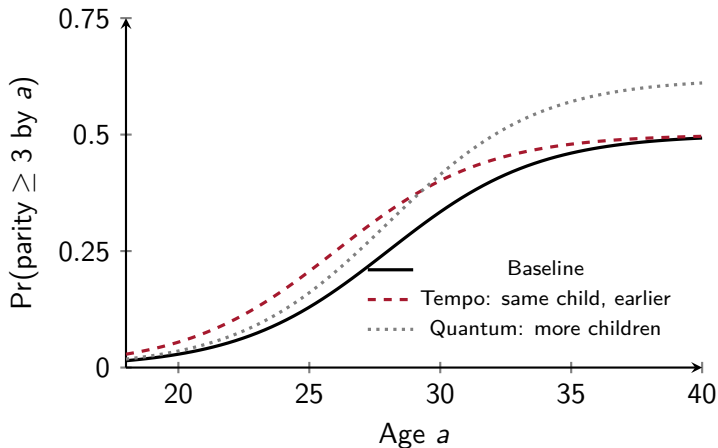
Appendix: Trend Differences Across NUTS-2 Regions



Each point is the pre-2012 linear trend in the share with three or more births by age 28, estimated separately for one NUTS-2 region, with a 95 percent interval. The joint test that all 26 slopes are equal rejects with $p < 0.0001$: regional fertility trends genuinely differ, which is why region-specific trends are needed.

[◀ Back](#)

Appendix: Tempo versus Quantum, Graphically



The hazard rises in both scenarios; only quantum raises the level at late ages. My hazard estimate is positive while the stock at fixed ages has not yet moved, which is the tempo pattern.

[◀ Back](#)

Appendix: All Subgroups

Subgroup	First birth	Second birth	Third birth	Fourth birth	Fifth birth
Turkish mother tongue	-0.120* (0.068)	0.199 (0.226)	0.220* (0.121)	0.094 (0.107)	0.025 (0.153)
Kurdish/Arabic mother tongue	-0.105 (0.114)	0.512* (0.296)	0.504** (0.256)	0.129 (0.153)	0.007 (0.230)
Lower education	-0.065 (0.155)	0.203 (0.174)	0.219 (0.136)	0.096 (0.096)	-0.038 (0.093)
Higher education	-0.063 (0.039)	0.089 (0.435)	0.490*** (0.169)	-0.017 (0.341)	-0.992 (1.275)
High-exposure border region	-0.262** (0.131)	0.497*** (0.164)	0.357*** (0.127)	0.025 (0.172)	0.037 (0.085)
Top exposure tercile	-0.153 (0.094)	0.007 (0.140)	0.396*** (0.110)	0.146 (0.136)	-0.117 (0.123)
Top exposure quintile	-0.161 (0.105)	0.017 (0.170)	0.392*** (0.140)	0.135 (0.162)	-0.250** (0.124)

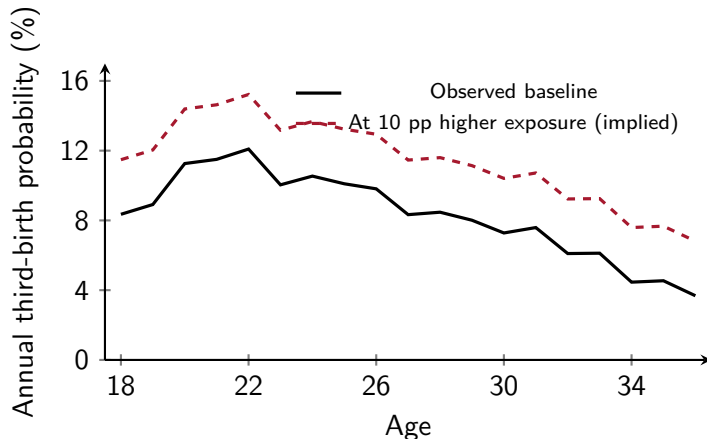
Bold: survives the BH correction within that subsample's five transitions ($q < 0.05$). The lower-education, tercile, and quintile placebo families are clean; Kurdish/Arabic, higher education, and the border region show differential pre-trends.

◀ Back

- I test many outcomes: five birth transitions, 26 cumulative cells, subsamples, and placebos. At the 5 percent level, some unadjusted rejections are expected by chance alone.
- The BH procedure controls the **false discovery rate**: the expected share of rejections that are false. The reported q -value is the smallest false discovery rate at which that cell rejects.
- q -values are computed within families of related tests, for example the five birth transitions.
- It uses only the unadjusted p -values, so anyone can recompute it from the tables; Anderson sharpened q -values agree throughout.

◀ Back

Appendix: The Main Result, Graphically



Weighted share of women with exactly two children who have a third at each age (2013/2018 person-years), and the same profile shifted by the 2SLS estimate at 10 percentage points more exposure: +3.1pp at every age, since the specification has no age interaction. On the 8.2 percent average baseline this is the 38 percent increase.

Appendix: Adding Age-Range-Specific Trends

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
<i>Panel A: Baseline (NUTS-2 and mother-tongue trends)</i>					
Refugee-to-native ratio R_{jt}	-0.070 (0.076)	0.235 (0.147)	0.313*** (0.106)	0.066 (0.099)	-0.048 (0.096)
BH q -value	[0.596]	[0.274]	[0.015]	[0.616]	[0.616]
<i>Panel B: Baseline plus age-range-specific trends</i>					
Refugee-to-native ratio R_{jt}	-0.060 (0.073)	0.234 (0.143)	0.332*** (0.101)	0.066 (0.098)	-0.041 (0.094)
BH q -value	[0.626]	[0.253]	[0.005]	[0.626]	[0.661]
Weighted transition mean	0.062	0.160	0.082	0.081	0.097
First-stage F (Panel A)	53.5	98.1	118.4	172.1	64.6
Person-years (Panel A)	163,980	51,359	51,677	25,161	10,806

Transition risk varies mechanically with age, so the hazard permits age-range-specific trends as a sensitivity; the third-birth estimate is essentially unchanged (0.332, $q = 0.005$).

◀ Back

Appendix: Cumulative Design Placebo

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.0010** (0.0005)	0.0015 (0.0014)	-0.0009 (0.0015)	-0.0009 (0.0009)	-0.0007 (0.0015)	-0.0021 (0.0023)
Parity ≥ 2	0.0003 (0.0003)	-0.0010 (0.0010)	0.0010 (0.0015)	0.0002 (0.0008)	-0.0021* (0.0012)	-0.0012 (0.0015)
Parity ≥ 3	0.0000 (0.0000)	-0.0002 (0.0004)	-0.0004 (0.0010)	-0.0007 (0.0015)	-0.0017 (0.0016)	-0.0067* (0.0036)
Parity ≥ 4	–	–	0.0004 (0.0006)	0.0022 (0.0020)	0.0028 (0.0042)	-0.0022 (0.0051)
Parity ≥ 5	–	–	-0.0001 (0.0003)	0.0010 (0.0010)	0.0037 (0.0025)	0.0050 (0.0045)

$B_j \times$ trend on pre-2012 cohort-age windows, one regression per cell. No corrected rejection across the 26 cells.

[◀ Back](#)

Appendix: Cumulative Design, OLS Counterpart

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	-0.10 (0.14)	-0.43 (0.56)	0.02 (1.64)	-0.67 (1.58)	-0.85 (1.11)	-0.97 (1.14)
Parity ≥ 2	0.03 (0.03)	0.16 (0.40)	-0.10 (0.91)	-0.66 (1.75)	-1.05 (2.40)	-1.02 (1.08)
Parity ≥ 3	0.00 (0.01)	0.11 (0.25)	-0.82 (0.49)	0.82 (1.42)	0.86 (1.78)	2.04 (1.66)
Parity ≥ 4	–	–	0.16 (0.32)	0.97 (0.79)	1.24 (1.44)	2.06 (2.24)
Parity ≥ 5	–	–	-0.10 (0.39)	0.70 (0.83)	2.84*** (0.90)	0.08 (1.72)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

OLS under the same NUTS-2 + mother-tongue specification as the 2SLS grid. The picture is unchanged: one isolated unadjusted rejection in 26 cells, nothing that survives correction — the null stock does not depend on instrumenting.

[◀ Back](#)

Appendix: Cumulative Third Birth Under the Next Two Trend Bases

Trend specification	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
NUTS-2 + MT (preferred)	0.00 (0.01)	0.30 (0.21)	-0.76 (0.76)	0.74 (1.76)	1.41 (2.81)	1.71 (1.96)
NUTS-1 + MT	0.00 (0.01)	0.45** (0.22)	-0.38 (0.59)	0.77 (1.67)	2.72 (2.68)	1.33 (1.77)
Region + MT	0.01 (0.01)	0.38 (0.25)	-0.37 (0.56)	1.25 (1.56)	2.89 (2.70)	1.06 (1.81)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Parity ≥ 3 by age, 2SLS. The stock stays null under all three bases — a single unadjusted age-20 blip under NUTS-1 that survives no correction — so the tempo-not-quantum conclusion does not hinge on the trend choice.

[Placebos under these trends](#)

[◀ Back](#)

Appendix: Cumulative Placebos Under the Next Two Trend Bases

Trend specification	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
NUTS-2 + MT (preferred)	0.0000 (0.0000)	-0.0002 (0.0004)	-0.0004 (0.0010)	-0.0007 (0.0015)	-0.0017 (0.0016)	-0.0067* (0.0036)
NUTS-1 + MT	0.0000 (0.0000)	0.0001 (0.0003)	0.0005 (0.0006)	0.0002 (0.0007)	-0.0000 (0.0007)	-0.0024 (0.0015)
Region + MT	0.0000 (0.0000)	-0.0000 (0.0003)	0.0004 (0.0007)	0.0000 (0.0009)	-0.0007 (0.0008)	-0.0024* (0.0014)
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Pre-2012 placebo, $B_j \times$ trend on the parity ≥ 3 row. Nothing rejects after correction under any of the three bases.

[Estimates under these trends](#)

[◀ Back](#)

Appendix: Hazard Estimates Under the Next Two Trend Bases

	NUTS-2 + MT (preferred)	NUTS-1 + MT	Region + MT
First birth	-0.070 (0.076) [0.596]	-0.048 (0.060) [0.552]	-0.041 (0.054) [0.644]
Second birth	0.235 (0.147) [0.274]	0.182 (0.143) [0.506]	0.201 (0.144) [0.403]
Third birth	0.313*** (0.106) [0.015]	0.266** (0.128) [0.191]	0.301** (0.129) [0.098]
Fourth birth	0.066 (0.099) [0.616]	0.049 (0.102) [0.631]	0.062 (0.096) [0.644]
Fifth birth	-0.048 (0.096) [0.616]	-0.070 (0.091) [0.552]	0.005 (0.090) [0.959]

2SLS coefficients, (clustered SEs), [BH q -values]. The third-birth estimate barely moves (0.27 to 0.31); no other transition turns significant under either coarser base.

Placebos under these trends

◀ Back

Appendix: Placebos Under the Next Two Trend Bases

	NUTS-2 + MT (preferred)	NUTS-1 + MT	Region + MT
First birth	0.0003 (0.0002) [0.259]	0.0002 (0.0002) [0.432]	0.0002 (0.0001) [0.299]
Second birth	-0.0002 (0.0003) [0.570]	-0.0001 (0.0002) [0.874]	-0.0002 (0.0002) [0.344]
Third birth	-0.0005 (0.0004) [0.259]	-0.0002 (0.0002) [0.432]	-0.0001 (0.0002) [0.642]
Fourth birth	0.0012 (0.0008) [0.259]	-0.0001 (0.0006) [0.926]	0.0000 (0.0005) [0.925]
Fifth birth	0.0011 (0.0008) [0.259]	0.0004 (0.0004) [0.432]	0.0006 (0.0004) [0.299]

Pre-2012 placebo slopes, $B_j \times \text{trend}$. No transition rejects after correction under any of the three bases, so the estimate does not depend on the one trend specification that happens to pass the placebo.

[Estimates under these trends](#) [◀ Back](#)

Appendix: Start-Year Sensitivity

	1987 (design)	1990	1995	2000
Third-birth hazard coefficient	0.313*** (0.106)	0.365*** (0.089)	0.391*** (0.109)	0.373*** (0.101)
First-stage F	118.4	119.6	120.4	125.3
Person-years	51,677	50,813	47,048	39,475

Later starts give 0.365 to 0.391; the design-based 1987 start is the most conservative.

[◀ Back](#)

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
Lagged ratio $R_{j,t-1}$	-0.094 (0.087) [0.467]	0.365* (0.209) [0.203]	0.517*** (0.129) [0.000]	0.007 (0.111) [0.947]	-0.041 (0.112) [0.889]
First-stage F	48.1	89.9	104.5	177.5	82.9
Person-years	160,357	50,903	51,470	25,059	10,768

A birth in year t reflects a conception decision around $t - 1$; lagged exposure fits sharper: the third-birth coefficient rises to 0.517 and every other transition stays null.

◀ Back

Appendix: First Stage and OLS Counterparts

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
$Z_{jt} (\times 10^{-5})$	2.579*** (0.353)	2.313*** (0.234)	2.296*** (0.211)	2.463*** (0.188)	2.802*** (0.349)
First-stage F	53.5	98.1	118.4	172.1	64.6
Person-years	163,980	51,359	51,677	25,161	10,806

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
Refugee-to-native ratio R_{jt}	-0.038 (0.047)	0.096 (0.126)	0.216* (0.120)	0.075 (0.168)	-0.151** (0.068)
Person-years	163,980	51,359	51,677	25,161	10,806

The OLS third-birth estimate (0.216) is smaller than 2SLS, consistent with refugee sorting toward lower-fertility, economically stronger provinces.

[◀ Back](#)

Appendix: Fixed 2011 vs. Annual Residence

	Fixed 2011 province	Annual province
<i>Panel A: 2SLS estimates</i>		
First birth	-0.070 (0.076)	-0.063 (0.082)
Second birth	0.235 (0.147)	0.206* (0.120)
Third birth	0.313*** (0.106)	0.387*** (0.123)
Fourth birth	0.066 (0.099)	0.100 (0.106)
Fifth birth	-0.048 (0.096)	-0.017 (0.106)
<i>Panel B: Pre-2012 placebo, $B_j \times$ trend</i>		
First birth	0.0003 (0.0002)	0.0002 (0.0002)
Second birth	-0.0002 (0.0003)	0.0001 (0.0002)
Third birth	-0.0005 (0.0004)	-0.0005 (0.0004)
Fourth birth	0.0012 (0.0008)	0.0011 (0.0008)
Fifth birth	0.0011 (0.0008)	0.0013 (0.0008)

Reconstructing each woman's province annually from her migration history gives 0.387 at the third birth; both definitions pass the placebo.

Appendix: Why Only the 2013 and 2018 Rounds?

The 2003 and 2008 rounds interviewed **only ever-married women** and lack the migration histories needed to assign pre-inflow residence. The ladder isolates each departure from the baseline design one step at a time:

	(1) Full sample 2011 province	(2) Ever married 2011 province	(3) Ever married childhood province	(4) Ever married, four waves, childhood province
First birth	-0.070 (0.076)	-0.180 (0.111)	-0.134 (0.116)	-0.222** (0.102)
Second birth	0.235 (0.147)	0.235 (0.147)	0.134 (0.140)	0.190 (0.143)
Third birth	0.313*** (0.106)	0.313*** (0.106)	0.164 (0.102)	0.352*** (0.118)
Fourth birth	0.066 (0.099)	0.066 (0.099)	0.078 (0.085)	0.137 (0.089)
Fifth birth	-0.048 (0.096)	-0.048 (0.096)	0.029 (0.098)	0.112 (0.087)
Third-birth person-years	51,677	51,677	51,677	101,943

Each column changes one thing: (1) is the baseline; (2) keeps only ever-married women; (3) assigns exposure at the childhood province instead of the 2011 province; (4) adds the 2003 and 2008 rounds.

⇒ **The earlier rounds are not worth their cost:** dropping never-married women changes nothing, and the older rounds force the noisier childhood-province assignment and fail their pre-period tests.

Four-wave estimates and placebos ◀ Back

Appendix: The Four-Wave Ever-Married Sample Fails Its Placebos

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
<i>Panel A: 2SLS estimates</i>					
Refugee-to-native ratio R_{jt}	-0.222** (0.102) [0.073]	0.190 (0.143) [0.198]	0.352*** (0.118) [0.014]	0.137 (0.089) [0.198]	0.112 (0.087) [0.198]
First-stage F	28.3	94.4	113.3	175.9	49.9
Person-years	218,950	96,086	101,943	53,539	26,232
<i>Panel B: Pre-trend placebo</i>					
$B_j \times \text{trend}$	0.0003** (0.0001) [0.052]	-0.0002 (0.0003) [0.439]	-0.0005** (0.0002) [0.037]	0.0008*** (0.0003) [0.019]	0.0006* (0.0003) [0.096]

Panel A: 2SLS on the ever-married sample of all four rounds (2003, 2008, 2013, 2018; childhood-province exposure); brackets are BH q -values. Panel B: its pre-period placebo rejects at four of five transitions after correction ($q < 0.10$) — differential pre-trends contaminate the added rounds, so the four-wave sample is corroboration only, never the design.

◀ Back

Appendix: Why Marriage Is Not a Headline Outcome

- **Design:** each birth risk set is defined by prior parity alone — a woman is at risk of birth p once she has $p - 1$ children, married or not — so no birth estimate depends on modeling marriage.
- **Evidence:** the marriage margin was estimated and does not respond.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever married by age a	-0.05 (0.22)	-0.57 (0.60)	0.51 (1.69)	-0.01 (1.05)	0.22 (0.92)	-0.30 (1.24)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Cumulative 2SLS on the ever-married indicator, same specification as the parity grid; no cell rejects, unadjusted or corrected. The discrete-time marriage hazard in the supplementary parity-progression panel is likewise null (0.040, SE 0.059), and so is the first birth — the transition closest to the marriage margin — in the main design.

[◀ Back](#)

$$W_{ij} = \alpha + \beta \bar{R}_j + \mathbf{X}_i' \Gamma + \delta_j + \mu_{t(i)} + \theta_{r(j)} t + \varepsilon_{ij}$$

- W_{ij} : the woman currently works (work-history module, ages 28 to 40 at interview, 2013 and 2018 rounds) or her husband worked in the last seven days (wife reported, 2008, 2013, and 2018 rounds).
- $\bar{R}_j$: average exposure at the woman's province, instrumented by the averaged shift-share; cross-sectional, one observation per woman.
- Unlike the fertility specification: macro-region trends $\theta_{r(j)} t$ and no mother-tongue trends, which would absorb the cross-group variation the mechanism works through.

◀ Back

Appendix: Subsample Pre-Period Tests

Subgroup	First birth	Second birth	Third birth	Fourth birth	Fifth birth
Turkish mother tongue	0.0002 (0.0002) [0.696]	-0.0004 (0.0003) [0.315]	-0.0001 (0.0004) [0.715]	0.0015* (0.0008) [0.315]	0.0003 (0.0008) [0.715]
Kurdish/Arabic mother tongue	0.0010*** (0.0003) [0.006]	0.0017* (0.0010) [0.112]	-0.0006 (0.0007) [0.449]	0.0021*** (0.0007) [0.006]	0.0035*** (0.0008) [0.001]
Lower education	0.0004 (0.0003) [0.233]	-0.0003 (0.0004) [0.481]	-0.0006 (0.0004) [0.233]	0.0013* (0.0007) [0.233]	0.0011 (0.0008) [0.233]
Higher education	-0.0004** (0.0001) [0.028]	-0.0000 (0.0010) [0.994]	-0.0024** (0.0010) [0.028]	-0.0056*** (0.0017) [0.009]	0.0041 (0.0142) [0.970]
High-exposure border region	0.0005** (0.0002) [0.080]	0.0004 (0.0009) [0.813]	-0.0007** (0.0003) [0.080]	-0.0001 (0.0003) [0.826]	0.0032*** (0.0007) [0.001]
Top exposure tercile	0.0003 (0.0002) [0.353]	-0.0002 (0.0003) [0.531]	-0.0004 (0.0004) [0.353]	0.0012 (0.0009) [0.353]	0.0014 (0.0008) [0.353]
Top exposure quintile	0.0003 (0.0002) [0.328]	-0.0004 (0.0003) [0.328]	-0.0004 (0.0004) [0.328]	0.0012 (0.0009) [0.328]	0.0017** (0.0007) [0.136]

Clean families: Turkish, lower education, tercile, quintile. Kurdish/Arabic, higher education, and the border region each reject at three transitions.

◀ Back

Appendix: Subgroup Third-Birth Estimates, OLS and Other Trends

	2SLS NUTS-2 + MT (preferred)	2SLS NUTS-1 + MT	2SLS Macro-region + MT	OLS NUTS-2 + MT
Turkish mother tongue	0.220* (0.121)	0.027 (0.165)	0.187 (0.172)	0.181* (0.095)
Kurdish/Arabic mother tongue	0.504** (0.256)	0.749** (0.307)	0.606*** (0.220)	0.225 (0.271)
Top exposure tercile	0.396*** (0.110)	0.365*** (0.139)	0.415*** (0.121)	0.245 (0.149)
Top exposure quintile	0.392*** (0.140)	0.357** (0.156)	0.434*** (0.133)	0.219 (0.174)

Third-birth hazard within each subsample under three trend bases and OLS. The tercile and quintile estimates are stable at 0.36 to 0.43 across all three 2SLS trend specifications; OLS is uniformly attenuated, as in the pooled design, consistent with refugee sorting toward lower-fertility provinces.

[◀ Back](#)

Appendix: Pooled Interaction Tests by Mother Tongue

	First birth	Second birth	Third birth	Fourth birth	Fifth birth
Effect, Turkish MT	-0.153** (0.072)	0.229 (0.209)	0.197 (0.132)	0.005 (0.125)	0.107 (0.159)
Effect, Kurdish/Arabic MT	0.121 (0.128)	0.199 (0.288)	0.537** (0.214)	0.160 (0.171)	-0.288** (0.146)
Difference (K/A – Turkish)	0.274** (0.120)	-0.030 (0.416)	0.340 (0.238)	0.156 (0.216)	-0.395* (0.237)
Unadjusted p	0.023	0.943	0.154	0.470	0.096
BH q -value	[0.115]	[0.943]	[0.256]	[0.588]	[0.240]
Placebo-difference BH q	[0.328]	[0.962]	[0.962]	[0.962]	[0.434]

Differences are right-signed at every transition but none survives the correction, which is why the gradient stays suggestive.

[◀ Back](#)

Appendix: Cumulative Grid, Kurdish/Arabic Subsample

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	-0.09 (0.26)	1.53 (1.64)	-5.60* (3.25)	-4.87 (3.47)	4.69 (3.13)	-1.22 (3.06)
Parity ≥ 2	-0.11 (0.11)	0.91 (1.24)	-0.88 (1.90)	-3.82 (3.54)	4.65 (3.43)	0.00 (1.71)
Parity ≥ 3	-0.02 (0.04)	0.29 (1.21)	-2.34 (1.47)	8.36*** (2.41)	1.35 (3.12)	-1.87 (2.02)
Parity ≥ 4	—	—	0.85 (1.36)	5.45** (2.68)	-3.58 (4.63)	2.74 (2.89)
Parity ≥ 5	—	—	-0.14 (1.02)	1.27 (3.10)	0.12 (4.33)	0.66 (3.37)

Parity 3 by 28 survives the subsample's own correction ($q = 0.014$); the subsample's placebo grid has rejections, so the cell is read with caution.

[◀ Back](#)

Appendix: Employment Estimates, OLS and Other Trends

	All women	Kurdish/ Arabic	Lower education	Border region
<i>Panel A: Woman currently working</i>				
2SLS, macro-region trends (preferred)	-2.651** (1.029)	-3.944** (1.674)	-4.790*** (1.268)	-2.199* (1.223)
2SLS, NUTS-1 trends	-3.609*** (1.270)	-5.352** (2.289)	-6.766*** (1.306)	-5.018** (2.005)
2SLS, NUTS-2 trends	-3.715* (2.011)	-6.827 (5.356)	-5.327* (2.814)	-5.940** (2.902)
OLS, macro-region trends	-2.246** (1.073)	-3.222** (1.342)	-3.366** (1.392)	-2.391 (1.425)
Observations	4,126	936	3,049	1,096
<i>Panel B: Husband worked in the last seven days</i>				
2SLS, macro-region trends (preferred)	2.013* (1.041)	-0.611 (1.482)	2.768** (1.250)	-0.814 (0.549)
2SLS, NUTS-1 trends	2.436** (1.039)	-0.335 (1.517)	3.332*** (1.236)	-0.539 (0.827)
2SLS, NUTS-2 trends	3.164** (1.483)	1.110 (1.612)	3.946** (1.612)	0.129 (1.540)
OLS, macro-region trends	0.731 (0.857)	-1.415 (0.944)	0.940 (0.864)	-0.698 (0.682)
Observations	12,760	2,869	10,076	3,659

The gender asymmetry — women's employment falls, husbands' does not — holds under NUTS-1 and NUTS-2 trends and in OLS; the female estimates grow under the finer trend bases while OLS is attenuated.

Appendix: Ideal Family Size, OLS and Other Trends

	2SLS Macro-region + MT (preferred)	2SLS NUTS-1 + MT	2SLS NUTS-2 + MT	OLS Macro-region + MT
Full sample (pooled)	1.197** (0.541)	0.992* (0.530)	2.111*** (0.707)	0.810* (0.448)
Lower education	1.533*** (0.549)	1.429** (0.572)	2.847*** (0.782)	1.179 (0.771)
Arabic/Kurdish mother tongue	3.034** (1.253)	2.878** (1.350)	4.655* (2.391)	2.128 (1.623)
High-exposure border region	1.566** (0.734)	1.216 (0.967)	1.633* (0.897)	0.827 (0.760)

Linear ideal family size under three trend bases and OLS. The estimate is positive in every cell and grows under the finest (NUTS-2 + mother-tongue) trends; OLS is attenuated but same-signed. The four-and-five-child threshold pattern likewise holds across specifications.

[◀ Back](#)

Beyond the sample-selection correction, the published Table 1 is not internally coherent:

- **The published “OLS” column is mechanically a 2SLS.** Six of its seven cells are numerically reproduced by the IV estimator (0.335 vs. 0.333; 2.536 vs. 2.466), while a genuine OLS gives 0.168; a real OLS sits near half its own IV for birth outcomes, the published column sits at parity.
- **No single instrument reproduces both headline cells.** The unweighted distance form fits the birth headline but overshoots number-of-children by 32 percent; the paper’s documented equation-(2) form fits number-of-children but overshoots the birth headline by 21 percent.
- **The reported first-stage F of 271 is not reproducible** from any standard estimator (cluster-robust ≈ 27 , robust $\approx 4,700$, Cragg-Donald $\approx 13,000$), and the paper’s own per-age F ’s vary 25-fold across adjacent age groups.

Full replication: 19 specification panels, three heterogeneity tables, and the independent TDHS re-run, adversarially verified.

[◀ Back](#)

Cell	Coef.	Conv. p	WCB p	WCB 95% CI	RI p	Prov.	N
Second-birth hazard	0.487	0.117	0.412	[-0.89, 1.05]	0.315	73	7,761
Third-birth hazard	0.478	0.082	0.451	[-1.02, 0.87]	0.153	69	8,406
Parity ≥ 3 by age 28 (cumulative)	8.028	0.005	0.174	[-5.27, 13.35]	0.078	68	1,951
Parity ≥ 4 by age 36 (cumulative)	2.568	0.320	0.414	[-6.36, 12.59]	0.520	58	1,055

Wild-cluster bootstrap and randomization inference, given exposure concentrated in few provinces. None of the cells survives, one more reason the gradient stays suggestive.

[◀ Back](#)